

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular/Supplementary Winter Examination – 2024**

**Course:** B.Tech.      **Branch :** Civil Engineering      **Semester :** V

**Subject Code & Name:** BTCVC502 Geotechnical Engineering

**Max.Marks:** 60

**Date:** 08/02/2025

**Duration:** 300.

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                                         | (Level/CO) | Marks     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                                                                  |            | <b>12</b> |
| 1           | Dry soil is a ----- phase system                                                                                                                                        | 1/1        | 1         |
|             | a. one      b. Two      c. Three      d. Four                                                                                                                           |            |           |
|             | The soil having particles of variety of sizes is said to be-----                                                                                                        | 1/1        | 1         |
|             | a. Well graded      b. Gap graded      c. Poorly graded      d. Uniformly graded                                                                                        |            |           |
|             | The specific gravity of a soil solids can be determined by using--                                                                                                      | 1/1        | 1         |
|             | a. Sieve      b. Pycnometer      c. Hydrometer      d. Oven                                                                                                             |            |           |
| 4           | The constant head permeability test is suitable for                                                                                                                     | 1/2        | 1         |
|             | a.CG soil      b. FG soil      c. CG & FG soil      d. None                                                                                                             |            |           |
| 5           | The critical exit gradient for the seepage of water out of a soil is                                                                                                    | 1/2        | 1         |
|             | a. $\frac{G-1}{1+e}$ b. $\frac{G-1}{1-e}$ c. $\frac{G+1}{1+e}$ d. $\frac{e-1}{1+e}$                                                                                     |            |           |
| 6           | A flow net is drawn for a weir. The total head loss is 8m. Number of Potential drops are 8 and the length of flow of path of last flow field is 1m.The exit gradient is | 1/2        | 1         |
|             | a. 0.5      b. 0.75      c. 1.0      d. 1.25                                                                                                                            |            |           |
|             | Deviatric pressure in triaxial cell is known as ----- pressure                                                                                                          | 1/3        | 1         |
|             | a.cell      b. chamber      c. Pore      d. Additional vertical                                                                                                         |            |           |
| 8           | A purely clayey soil is having unconfined compressive strength of 40kpa, its cohesion will be -----kPa                                                                  | 1/3        | 1         |
|             | a.10      b. 20      c. 40      d. 80                                                                                                                                   |            |           |
| 9           | Remoulded strength of soil having undisturbed strength 50ka and sensitivity of 2 is --- kPa                                                                             | 1/3        | 1         |

|      |                                                                                                                                                                                                                                                                                                                                                         |                 |              |                                       |     |     |     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------|---------------------------------------|-----|-----|-----|
|      | a. 25                                                                                                                                                                                                                                                                                                                                                   | b. 50           | c. 75        | d. 100                                |     |     |     |
| 10   | In a compaction test, as a compaction effort is increased, the optimum water content                                                                                                                                                                                                                                                                    |                 |              |                                       | 1/4 | 1   |     |
|      | a. Decreases                                                                                                                                                                                                                                                                                                                                            | b. Remains same | c. Increases | d. Increases first and then decreases |     |     |     |
| 11   | In case of compaction, Dry density of soil is maximum when water content of soil is                                                                                                                                                                                                                                                                     |                 |              |                                       | 1/4 | 1   |     |
|      | a. 0 %                                                                                                                                                                                                                                                                                                                                                  | b. 100 %        | c. OMC       | d. None                               |     |     |     |
| 12   | Coefficient of compressibility is the slope of virgin portion of -----curve                                                                                                                                                                                                                                                                             |                 |              |                                       | 1/4 | 1   |     |
|      | a. flow                                                                                                                                                                                                                                                                                                                                                 | b. compaction   | c. e - p     | d. e - logp                           |     |     |     |
|      |                                                                                                                                                                                                                                                                                                                                                         |                 |              |                                       |     |     |     |
| Q. 2 | Solve the following.                                                                                                                                                                                                                                                                                                                                    |                 |              |                                       |     | 12  |     |
| A)   | Derive relation $\gamma_d = \frac{G_{yw}}{(1+e)}$ use 3 phase labelled diagram.                                                                                                                                                                                                                                                                         |                 |              |                                       | 5/1 | 6   |     |
| B)   | Following observations were recorded during specific gravity test Find Specific gravity of soil.<br>Mass of pycnometer = 40.1gm<br>Mass of pycnometer and dry soil = 65.8gm<br>Mass of pycnometer dry soil and water = 154.5gm<br>Mass of pycnometer and water = 138.5gm                                                                                |                 |              |                                       | 3/1 | 6   |     |
|      |                                                                                                                                                                                                                                                                                                                                                         |                 |              |                                       |     |     |     |
| Q. 3 | Solve the following.                                                                                                                                                                                                                                                                                                                                    |                 |              |                                       |     | 12  |     |
| A)   | With a neat labelled sketch explain plasticity chart. Explain meaning of each dual symbol used to classify the soil (Minimum six symbols)                                                                                                                                                                                                               |                 |              |                                       | 4/2 | 6   |     |
| B)   | A falling head permeability tests was carried out on a 120cm long sample of silty clay. The diameter of sample and the stand pipe were 10cm and 1.0cm, respectively. the water level in stand pipe was observed to fall from 80 to 60cm in 12 min, determine the coefficient of permeability of the soil in m/day                                       |                 |              |                                       | 4/2 | 6   |     |
|      |                                                                                                                                                                                                                                                                                                                                                         |                 |              |                                       |     |     |     |
| Q. 4 | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                         |                 |              |                                       |     | 12  |     |
| A)   | Find the average coefficient of permeability soil mass made up of three horizontal layers. The first and second layers have the same thickness of 0.6m each. The third layer is 0.7m thick the coefficient of permeability of first, second, and third layers are $1 \times 10^{-3}$ , $2.5 \times 10^{-2}$ and $4.5 \times 10^{-3}$ cm/s, respectively |                 |              |                                       | 4/2 | 6   |     |
| B)   | Direct shear test was conducted on a river sand using sample of size 60mm X 60mm X 25mm and following results were obtained determine shear parameters of soil                                                                                                                                                                                          |                 |              |                                       |     | 6   |     |
|      | Normal force(N)                                                                                                                                                                                                                                                                                                                                         | 80              | 160          | 240                                   | 320 | 400 | 480 |
|      | Shear force (N)                                                                                                                                                                                                                                                                                                                                         | 50              | 101          | 149                                   | 201 | 248 | 302 |

|             |                                                                                                                                                                                                                                                                                                                                                                |     |           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|
|             |                                                                                                                                                                                                                                                                                                                                                                |     |           |
| C)          | Compare direct shear test and tri-axial shear test (minimum 6 points)                                                                                                                                                                                                                                                                                          | 4/2 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                |     |           |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                         |     | <b>12</b> |
| A)          | What is compaction of soil? Discuss with neat illustrative sketches factors affecting compaction of soil.                                                                                                                                                                                                                                                      | 4/3 | 6         |
| B)          | Draw $e - p$ curve for soil? Label different branches of this curve. What are the slopes of these branches?                                                                                                                                                                                                                                                    | 4/4 | 6         |
| C)          | Calculate total active earth pressure and its position with respect to bottom of wall acting on a retaining wall of height 9m retaining two layered soil on back side of it.<br>Top layer 4.2m thick $\gamma = 19 \text{ kN/m}^3$ , $c = 0$ and $\phi = 30^\circ$ followed by second layer having $\gamma = 18 \text{ kN/m}^3$ , $c = 0$ and $\phi = 27^\circ$ | 5/4 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                |     |           |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                         |     | <b>12</b> |
| A)          | The following data refer to a sample of soil:<br>Percent passing 4.75 mm IS Sieve = 64<br>Percent passing 75- $\mu$ IS Sieve = 6<br>Uniformity Coefficient = 7.5<br>Coefficient of Curvature = 2.7<br>Plasticity index = 2.5<br>Classify the soil. Draw flowchart of classification to classify soil.                                                          | 5/2 | 6         |
| B)          | Explain Rehmann method to calculate active earth pressure acting on wall with the help of illustrative sketch (use mathematical instruments to draw sketch – avoid freehand sketch)                                                                                                                                                                            | 5/4 | 6         |
| C)          | Draw typical grain size distribution curve for well graded soil. How do you calculate coefficient of curvature and coefficient of uniformity for the soil. What are the values of $C_c$ and $C_u$ for poorly graded soil?                                                                                                                                      | 5/1 | 6         |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                             |     |           |